

Why the Model 686 Arbitrary Waveform Generator Is the Go-To Tool for Semiconductor Test Engineers

20 GS/s sample rate, 14-bit resolution, and 120 ps edges give test engineers one instrument for high-speed digital, analog, and mixed-signal characterization.

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SAN RAFAEL, Calif., August 25, 2026. When a semiconductor device fails a test bench and passes in the field, the instrument generating the stimulus is usually the first thing test engineers question. Precision and speed are non-negotiable when validating today's high-speed devices, and the Berkeley Nucleonics Model 686 arbitrary waveform generator (AWG) has become a fixture on mixed-signal test benches for exactly that reason: it replicates real-world signal conditions with the fidelity and control semiconductor validation demands.

The Model 686 combines a 20 GS/s real-time sample rate with true 14-bit vertical resolution, a pairing that lets engineers emulate the fast, high-resolution signals modern devices actually see in operation — without the distortion that creeps in when an instrument trades resolution for speed, or speed for resolution.

Speed and Resolution Without Compromise

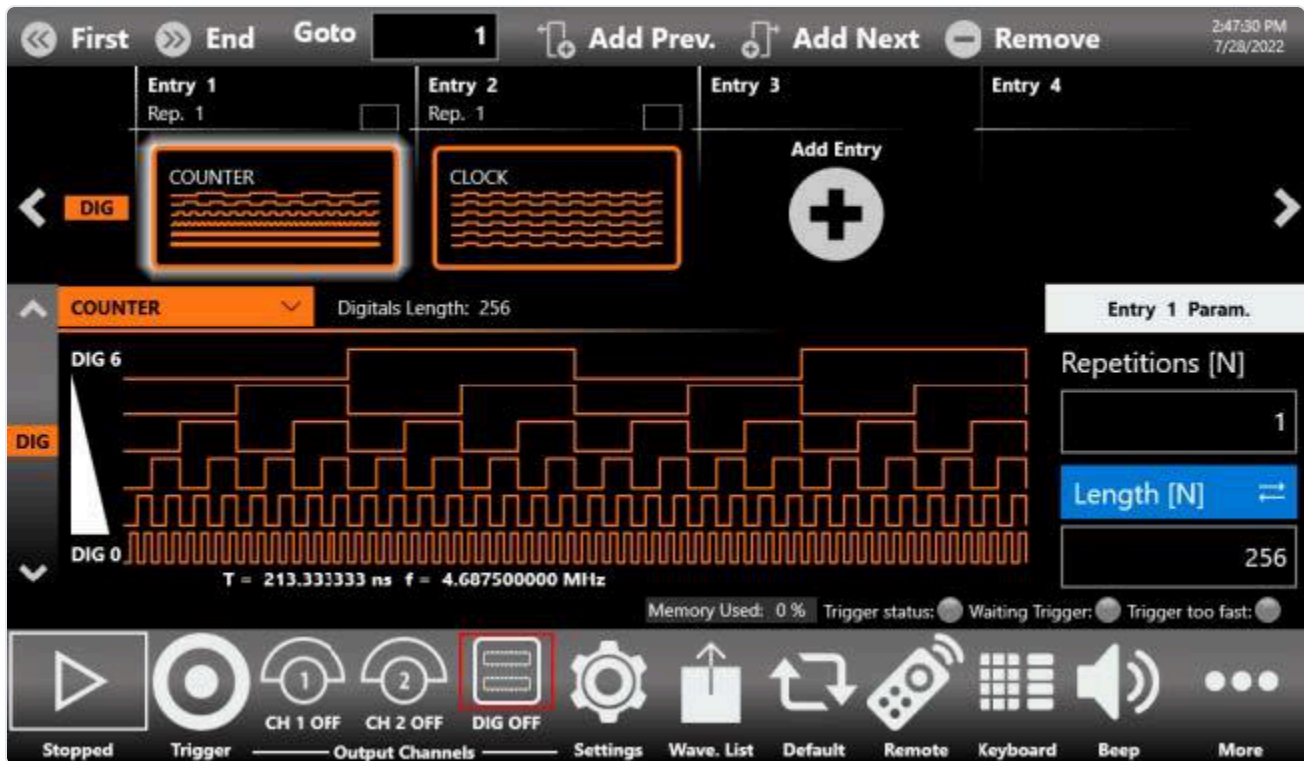
Sample rate is programmable from 1 S/s up to 20 GS/s, with a measured 3 dB bandwidth of 5.8 GHz (sin(x)/x compensated). That combination of speed and 14-bit resolution keeps generated waveforms clean and low-distortion, so the signal a device under test sees on the bench matches what it will see in the field — critical for engineers characterizing how a part actually behaves rather than how it behaves under an idealized stimulus.

Sharp Edges for Sharp Devices

Semiconductor testing lives and dies on edge quality. The Model 686 delivers a square-wave rise/fall time of 120 ps (10% to 90%), with output amplitude up to 5 Vpp at frequencies to 3.5 GHz, tapering to 3 Vpp as frequency climbs to 6.5 GHz. That gives test engineers both the sharp transitions needed for fast-switching digital circuits and the signal strength needed for analog front-end characterization — from the same channel, without re-cabling to a second instrument.

Built for Serial and Parallel Digital Test

The optional Pulse Pattern Generator (PPG) mode is purpose-built for digital and mixed-signal semiconductor work. It generates PRBS and multi-level PAM patterns, plus custom patterns up to 12 MSymbols, at pattern rates up to 6.5 Gbaud, with digital outputs delivering up to 10 Gb/s per line. Paired with up to 32 digital lines running in tight synchronization with the analog channels, the Model 686 covers mixed-signal test scenarios — a baseband analog signal alongside toggling control lines and parallel data — from a single chassis, with everything aligned in time.



The Simple Rider PPG mode interface, used to build PRBS and custom digital test patterns directly on the instrument's touchscreen.

Room for Complex Test Sequences

Test scenarios are rarely simple. The Model 686 offers up to 9 Gpts of waveform memory per channel in Full Rate mode, enough to load extensive waveform data, multi-step test sequences, or captured real-world signals without truncating a test. As requirements grow, up to four Model 686 units can be linked over a dedicated intra-chassis synchronization bus to form a single 16-analog, 128-digital-channel system without losing timing accuracy — useful for parallel test stations or large-scale IC evaluation where one chassis runs out of channels.

“Semiconductor test benches need instruments that don't lie — clean edges, accurate amplitude, and enough memory to hold a full test sequence without shortcuts,” said Amira Sekkouty, Associate Application Engineer at Berkeley Nucleonics. “The Model 686 gives engineers one box that covers the function-generator, arbitrary-waveform, and pattern-generation modes a mixed-signal test bench actually uses, instead of three instruments that each do one job.”

The included Simple Rider software runs all three operating modes — AFG (DDS function generator), True Arb (variable-clock arbitrary waveform), and PPG (pattern generator) — from a touchscreen-

ready interface, so engineers can move between waveform creation, sequencing, and pattern generation without switching software.

Beyond the Semiconductor Bench

While semiconductor validation is where the Model 686 sees its heaviest use, the same speed, resolution, and channel count make it a regular fixture in photonics, aerospace, and RF labs — a versatility that makes it a durable investment rather than a single-purpose tool.

Availability

The Model 686 is available through Berkeley Nucleonics and its representatives worldwide. Full specifications, configuration options, and quote requests are on the [Model 686 datasheet page](#).

About Berkeley Nucleonics

Berkeley Nucleonics Corporation (BNC) has been a trusted name in precision electronic instrumentation since 1963, serving defense, research, academic, and industrial customers with digital delay generators, arbitrary waveform generators, RF/microwave signal generators, pulsed power systems, and nuclear radiation detection equipment. With 60+ years of reliability and responsive technical support, BNC is flexible and listens to your application.

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